arrived late frequently, were unavailable, or required fallback assumptions were identified as reducing the accuracy of authorization decisions.

For example, ACME found that device posture signals for managed laptops were not always up to date. Some devices only reported posture at login, which meant a device could become non-compliant hours later without the policy engine noticing. By tightening posture refresh intervals and enforcing expiration rules on outdated signals, ACME improved the reliability of its device-trust data. Similar updates were made for authentication freshness and operational signals, such as on-call status and approval records.

As signal quality improved, the policy engine could make more accurate decisions with fewer false denials and exceptions. High-quality signals reduced the need for conservative fallback rules and helped ensure that access decisions truly reflected ACME's intent rather than compensating for missing or unreliable data.

15.5.3 Decision consistency

Decision consistency reflects whether authorization behaves predictably across systems when the same conditions apply. Before adopting PBAC, ACME often saw identical requests produce different outcomes depending on which application processed them. Each system encoded its own assumptions about roles, risk, and trust, sometimes in configuration files, sometimes directly in code. Consequently, a principal could be granted access in one system and denied in another, even if nothing about the identity, device posture, or context changed.

ACME assessed decision consistency by running standardized authorization scenarios across multiple systems and comparing outcomes. The authorization team crafted a set of canonical test requests—combinations of principal identity, attributes, device posture, time of day, and operational context—and examined how various applications responded to these requests. Early results showed notable variation, particularly in systems that still depended on embedded authorization logic or network-based shortcuts.

As foundational and shared operational policies became more widely adopted, these discrepancies began to disappear. For example, ACME tested